

BASEBALL ANALYSIS REPORT

Prepared By:

Jeffrey Flores

Prepared For:

Professor Li,
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I. Summary

The first question I was trying to answer with my analysis of the baseball dataset was if a relationship existed between wins and on-base percentage (OBP). OBP is a metric in baseball used to calculate how frequent a hitter gets on base per plate appearance. Based on my regression analysis of both quantitative variables I was able to find a relationship between the two. This means that wins can be explained by on-base percentage. The second question I was trying to find an answer to was if batting average (BA) and a team making the playoffs had a relationship. Batting average is a baseball metric to measure a hitter's success at the plate. After running an analysis of variance and correlation test on the categorical variable, playoff appearance, and quantitative variable, BA, I was able to find a relationship. This means that batting average can be explained by a team making the playoffs.

II. Data Description

The data I used for this analysis came from a baseball dataset containing 15 different variables. It has 50 years of data, spanning from 1962 to 2012. The data contains baseball metrics such as runs scored, runs allowed, slugging percentage, teams, and much more. Although I won't go into detail into every single variable, I will give some context on the variables I used for this analysis. As I previously mentioned, I used variables batting average, wins, on-base percentage, and if a team made the playoffs. All of these variables were quantitative except for the playoff appearance variable which was a categorical variable. It's important to note that the dataset contains data about the whole team. For example, metrics such as batting average, on-base percentage, slugging percentage usually pertain to a singular hitter. The dataset instead records a team's batting average, slugging percentage and on-base percentage.

The first variable I will be talking about is batting average, which is a measure of a hitter's success during a plate appearance. It is calculated by dividing the number of hits by at-bats of a hitter and is usually reported to three decimal places. The next variable, wins, is simply recorded by the amount of wins a team has in a single 162-game season. On-base percentage is a metric which shows how frequent a hitter reaches on base per plate appearance. It is calculated by adding the hits, walks, and hit by pitches of a hitter then dividing it by the sum of

their walks, at-bats, hit by pitches and sacrificial flies. Like batting average, it's usually reported to three decimal places. Finally, playoff appearance is simply recorded by a 0 or 1 where 1 signifies a team making the playoffs and 0 meaning they didn't.

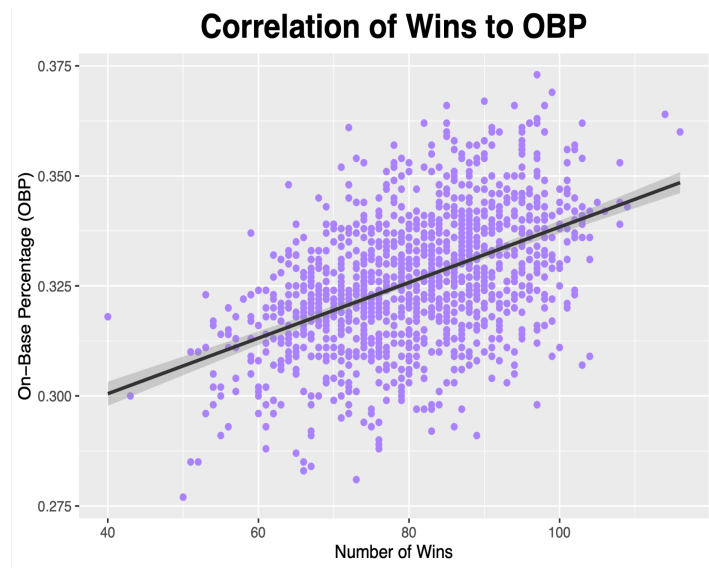
III. Data Manipulation

In order for my analysis on the categorical variable, playoff appearance, to work I had to convert it from an integer variable to a factor variable. The reason being that the results of playoff appearance in the baseball dataset were just integers of 0 and 1. Therefore I had to change this in order to format my two groups. I was able to manipulate this data via the `as.factor` function in R. With the help of this function, it let me categorize my playoff appearance results into two different groups. After I did this, I then changed the levels of the playoff variable so they could be informative, as the levels 0 or 1 didn't really mean much. I converted the 0 level to no meaning the team didn't reach the playoffs and the 1 to yes signifying a team reached the playoffs.

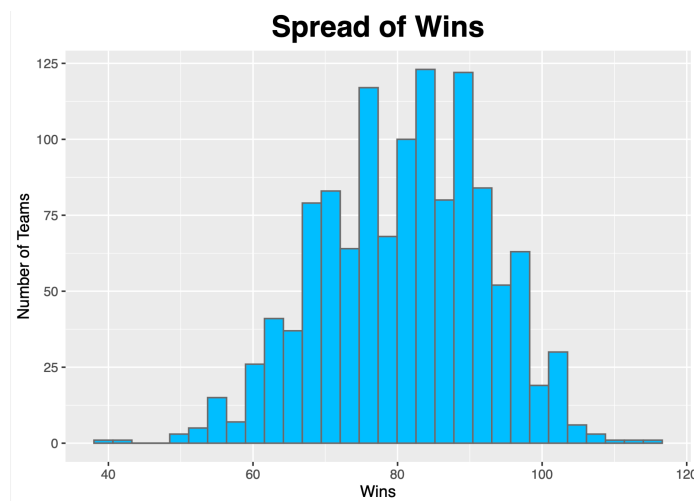
Another way I manipulated the data was through subsetting. After I was able to find a relationship between batting average and playoff appearance, I wanted to find more about their correlation. Specifically, I wanted to compare the min, median, and max batting average from playoff and non-playoff teams. Therefore, I subsetting the baseball dataset to only include playoff teams and non-playoff teams. Then selected year, team, and batting average as the variables I wanted to observe. After I did this, I simply used the summary function on both datasets to find the information on batting average I was interested in. Once I was able to find the min, max, and median of both datasets I then subsetting these two different baseball datasets to see which team reached these marks and in what year.

IV. Data Analysis

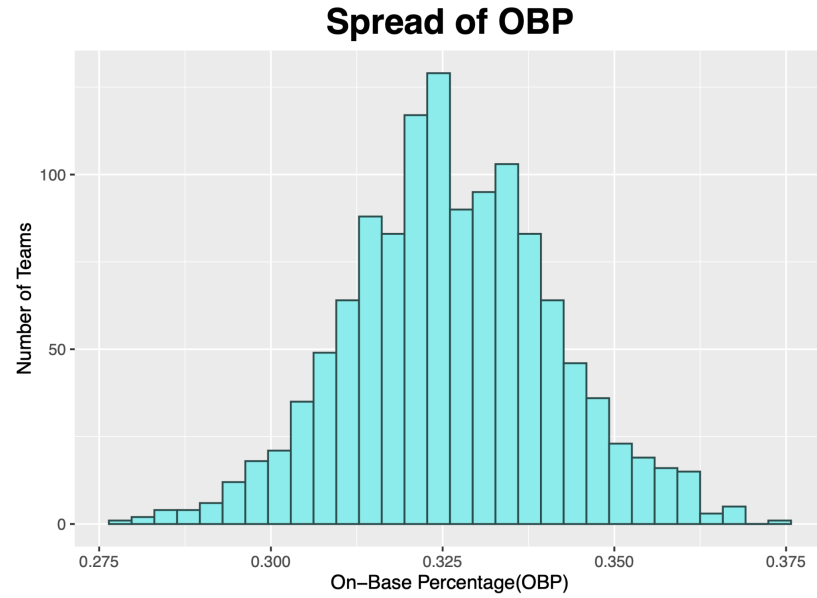
A. Data Visualization



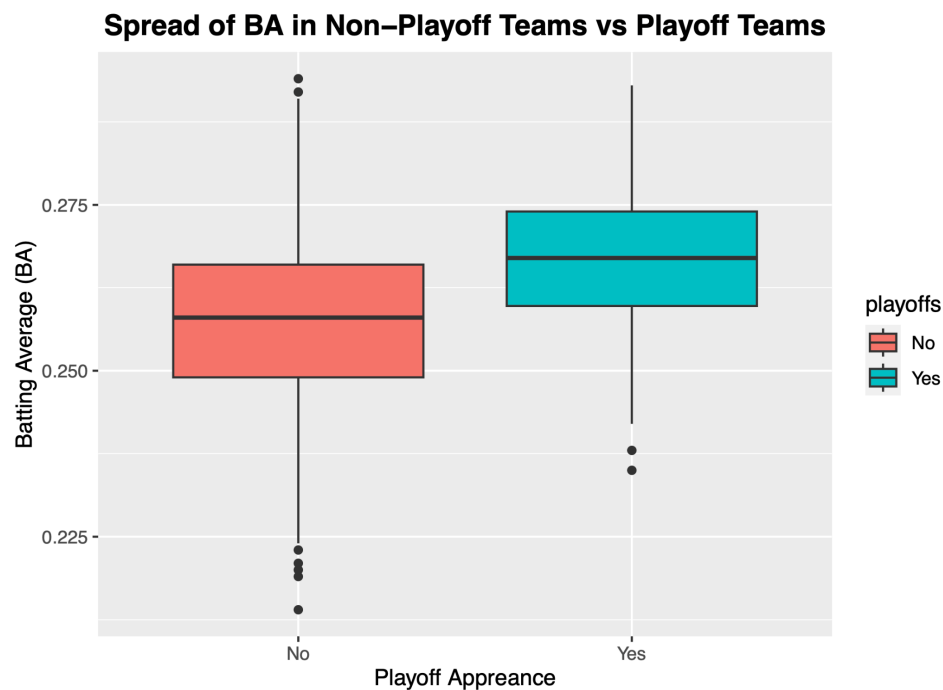
The scatter plot pictured above shows the relationship between the number of wins and on-base percentage. I produced this graph in order to see if any type of correlation existed. As you can see, there seems to be a moderate positive correlation between the two variables. Therefore, I continued with building my statistical model.



The histogram above shows the distribution of wins throughout the 50 year span of the data. I produced this histogram because I wanted to see if the data followed a normal distribution. Based on this graph and a Shapiro test I ran on this variable, I was able to see that the data was not normal. Therefore when I ran my correlation test with this variable, I had to make sure it was a Spearman correlation test.



The graph above shows the distribution of the on-base percentage variable. As you can see, the variable does seem to follow a normal distribution which was further proven to be right once I ran a Shapiro test.



The boxplot above shows the difference in batting average between playoff and non-playoff teams. I produced the graph to see if some sort of relationship existed. As you can see there seems to be some type of relationship, therefore I decided to continue building my statistical model.

B. Data Summary

Since the dataset contained 15 different variables I will not talk about the summary statistics about every single variable. Instead, I would like to focus on variables which I think are important to mention. The first variable which I will be talking about will be wins. The median amount of wins throughout the 50 years of recorded baseball was 81, with the max being 116 and a min of 40. For on-base percentage, a team's median in 50 years was 0.337, with a max of 0.3730 and a min of 0.277. When talking about batting average, a team's median batting average throughout 50 years was 0.260, with a max of 0.294 and a min of 0.214. For runs allowed, a team allowed a median of 709 runs per season throughout 50 years, a maximum of 1009 and a minimum of 472 runs.

C. Model Building

When answering if a relationship existed between wins and on-base percentage, I knew I had to create a regression model. Throughout the process of building the model, I encountered no issues. The only thing I was worried about was the wins data not being normally distributed. But this caused no problem within my regression model. When building my analysis of variance (ANOVA) model for the relationship between playoff appearance and batting average I encountered a problem. The problem in question was that the homogeneity of variance assumption for an ANOVA model was violated. This is a big issue as it lowers the accuracy of my model and questions the validity of my results. Since an ANOVA model is used for more than two groups and I used it for only two groups, this could have caused the issue. As a result of this issue I had to run a Kruskal and correlation test to see if a relationship existed.

D. Final Model

For my linear regression model between wins and on-base percentage, my response variable was wins while the predictor variable was on-base percentage. I built it this way as I wanted to see if wins can be explained by batting average. When examining the relationship between playoff appearance and batting average, the first thing I did was transform the integer variable, playoff appearance, to a factor variable. This helped me create the two groups I wanted to investigate. Once the groups were formed, I built my ANOVA model. This is where I realized an issue. As previously said, my model was not valid as one of the assumptions

was not met. Therefore, I had to go another route to find out if any sort of correlation existed between playoff appearance and batting average. The first thing I did was run a Kruskal test on the variables, which is an alternative to an ANOVA model and is used when the assumptions of an ANOVA model are not met. The problem with the Kruskal test is that it is often used when dealing with a categorical variable with more than two groups. Since my categorical variable had only two groups, yes and no, I wanted to run a point biserial correlation just to make sure my results were valid.

E. Model Fit

In terms of my regression model I was satisfied, as all the assumptions of a regression were met. I was able to figure this out by looking at the different plots of the model. For the linearity assumption, I looked at the residuals vs. fitted plot. Where I saw the residuals being spread equally around the horizontal line without any distinct pattern. This is an indicator of the linearity assumption being met. To check the homoscedasticity of my residuals, I looked at the scale location plot. Once viewing the plot you can see the residuals are spread randomly with equal spread of points around the horizontal line. Therefore the homoscedasticity assumption was met. For the normality assumption, I looked at the QQ plot and saw the residuals followed a straight line, therefore the assumption was fulfilled. In terms of potential pitfalls for this model, I'm worried about the multiple R-squared coefficient. I will go into further detail of this coefficient during the model interpretation section.

When talking about my ANOVA model between playoff appearance and batting average, I was not satisfied. The reason being that the homoscedasticity of variance assumption was not met. I was able to figure this out by running a Levene and Bartlett test on both variables. Where I saw the p-value of both tests being less than 0.05 meaning there is evidence suggesting my variance across groups is statistically different. This is a pitfall as it could invalidate my results, so I decided to run other tests to figure out if a relationship existed.

F. Model Interpretation

The first model I will be talking about is the regression model. When looking at the residual standard error, you usually want it to be as small as possible because it is an indicator the regression model is close to the actual values. In my regression model between wins and on-base percentage the residual error was 10.04 which is low as there are 162 games in a single

baseball season. The next coefficient I will be talking about is the multiple R-squared coefficient. Its value of 23.22% means that OBP can only describe some of the variation in wins,. therefore it could mean the model might not be fitting the data well. When talking about the p-value of my regression model it is extremely small. This shows the variables are statistically significant, meaning a relationship does exist. When talking about my ANOVA model for batting average and playoff appearance, the p-value was extremely low meaning there is statistical difference between the two groups. This was a good result but as soon as I found out one of the assumptions were not met, I ran a point biserial correlation test. There I also saw the p-value was also extremely low, therefore I concluded a relationship does exist between the two variables.

V. Conclusion

Overall, with the help of my regression model I was able to observe that a relationship did exist between the two quantitative variables on-base percentage (OBP) and wins. This means that wins can be explained by OBP. The implication of this relationship signifies that teams should strive to increase their on-base percentage as it helps in increasing a team's number of wins. When investigating the categorical variable, playoff appearance, and quantitative variable batting average (BA), I concluded a relationship did exist. I was able to find this result through a Kruskal and correlation test. This was not my original idea, as I previously planned to construct an ANOVA model but this didn't end up working. The implication of this relationship shows that teams should work on increasing their batting average, as high batting average indicates a playoff appearance. To further analyze the relationship, I would like to run an analysis on more observations. Although the dataset does contain 50 years of data, collecting more data will further add validity to my conclusions. Finally, for the simple regression model relationship, I could perhaps add more variables to construct a multiple regression model to see what other variables affect wins in baseball. This will give me an accurate idea on what teams should work on if they want to win.

Appendix

Note: The R-script can be found within the submission.